阿思丹 | ASDAN China

MATH KANGAROO CHINA

Level 6

Year 11 & Year 12

考试说明 / Instruction

Time allowed: 75 minutes
考试时长：75 分钟
Language: English & Chinese
试卷语言：英文和中文

This is a 30-question multiple-choice competition.
考试包含 30 道单项选择题。
Problems 1 to 10 are worth 3 points each.
1-10 题每题 3 分
Problems 11 to 20 are worth 4 points each.
11-20 题每题 4 分
Problems 21 to 30 are worth 5 points each.
21-30 题每题 5 分
1 point will be deducted for an incorrect answer and no point for a blank answer.
答错扣 1 分，不答不扣分。
The full score is 150 (Starting point is 30)
总分为 150 分（起始分数为 30 分）

Do not open this booklet until you are told to do so.
收到监考老师指令后，方可打开试卷册。
Mark your answers clearly on the answer sheet using a **2B pencil.**
务必使用 **2B 铅笔**在答题卡上正确填涂自己的答案。
Calculator cannot be used during the competition.
考试期间禁止使用计算器。

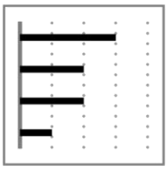
Good Luck!

预祝考试顺利

3 points

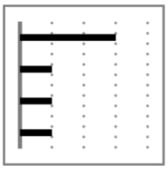
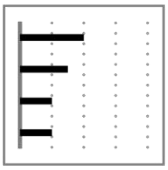
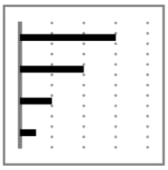
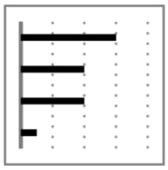
1. On Henry's smartphone, the diagram shows how much time he spent last week on each of his apps.

下图表示 Henry 上周在每个手机应用程序上的使用时长。



The apps are ordered from greatest to least time spent. This week, he spent exactly the same amount of time as last week on two of his apps, but only half as much time on the other two. Which of the diagrams below cannot be the diagram for this week?

应用程序按照使用时间最长至最短的顺序依序排列。这周，他花在两个应用上的时间和上周一样长，但花在另外两个应用上的时间都只有上周的一半。请问下面哪张图不能表示本周的使用时长?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

2. How many positive three-digit integers are divisible by 13?

请问有多少个三位正整数可以被 13 整除?

(**A**) 68 (**B**) 69 (**C**) 70 (**D**) 76 (**E**) 77

3. Bella is older than Charlie and younger than Lily. Teddy is older than Bella. Which two people could be the same age?

Bella 的年龄比 Charlie大，但比 Lily 小，Teddy 比 Bella 大。请问哪两个人的年龄可能一样大?

(**A**) Charlie and Teddy Charlie 和 Teddy

(**B**) Teddy and Lily Teddy 和 Lily

(**C**) Lily and Charlie Lily 和 Charlie

(**D**) Bella and Lily Bella 和 Lily

(**E**) Teddy and Bella Teddy 和 Bella

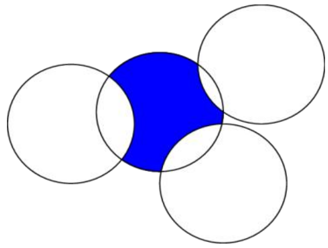
4. The product of the digits of a 10-digit integer is 15. What is the sum of the digits of this number?

有一个十位整数，其各个数位上的数字之积为 15，请问它的各个数位上的数字之和是多少?

(**A**) 8 (**B**) 12 (**C**) 15 (**D**) 16 (**E**) 20

5. Four circles, each of radius 1, intersect as shown. What is the perimeter of the shaded region?

四个半径均为 1 的圆按照如图所示的方式相交，请问阴影区域的周长是多少?



(**A**) π (**B**) Some number between $\frac{3\pi}{2}$ and 2π 介于 $\frac{3\pi}{2}$ 和 2π 之间

(**C**) $\frac{3\pi}{2}$ (**D**) 2π (**E**) π^2

6. David writes, in increasing order, all the integers from 2 to 2022 which use only 0s and 2s. What is the number in the middle of his list?

David 按照从小到大的顺序写下 2 到 2022 之间只由 0 和 2 组成的全部整数，请问位于中间的数是多少?

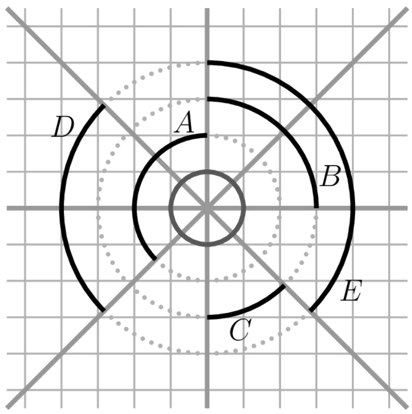
(**A**) 200 (**B**) 220 (**C**) 222 (**D**) 2000 (**E**) 2002

7. How many real solutions does the equation $(x - 2)^2 + (x + 2)^2 = 0$ have?
方程 $(x - 2)^2 + (x + 2)^2 = 0$ 有几个实数解?

(**A**) 0 (**B**) 1 (**C**) 2 (**D**) 3 (**E**) 4

8. Four lines intersect forming eight equal angles. Which black arc has the same length as the small grey circle?

四条直线相交形成八个相等的角。请问哪条黑色弧线的长度与最里面灰色小圆的周长相等?



(**A**) A (**B**) B (**C**) C

(**D**) D (**E**) E

9. Let a, b, c be non-zero numbers. The numbers $-2a^4b^3c^2$ and $3a^3b^5c^{-4}$ have the same sign. Which of the following is definitely true?

已知 a、b、c 都是非零数字，$-2a^4b^3c^2$ 和 $3a^3b^5c^{-4}$ 的符号相同。请问下列哪项肯定是正确的?

(**A**) $ab > 0$ (**B**) $b < 0$ (**C**) $c > 0$ (**D**) $bc > 0$ (**E**) $a < 0$

10. Mike has marked the points A, B, C and D in this order on a straight line, as shown in the diagram.

Mike 在一条直线上依次标出 A、B、C 和 D 四个点，如下图所示。

The distance between A and C is 12cm and between B and D, 18cm. What is the distance between the midpoint of AB and the midpoint of CD ?

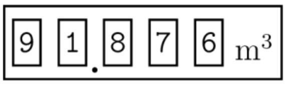
点 A 和点 C 之间相距 12cm，点 B 和点 D 之间相距 18cm。请问 AB 的中点和 CD 的中点之间相距多少？

(**A**) 15cm (**B**) 12cm (**C**) 18cm (**D**) 6cm (**E**) 9cm

4 points

11. When he looks at the water meter in his bathroom, Tony notices that all the digits on the meter are different.

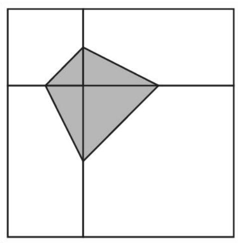
Tony 查看浴室的水表时发现水表上每个数位上的数字都不一样。



How much water will be used until the next time all the digits on the meter are different?
请问水表上各个数位上的数字再次不一样时，已经用完了多少体积的水？

(**A**) 0.006m^3 (**B**) 0.034m^3 (**C**) 0.086m^3 (**D**) 0.137m^3 (**E**) 1.048m^3

12. A large square is divided into two unequal squares and two equal rectangles, as shown. The vertices of the shaded quadrilateral are the midpoints of the sides of the two squares. The area of the shaded quadrilateral is 3. What is the area of the unshaded part of the large square?



将一个大正方形划分为两个大小不同的小正方形和两个完全相同的小矩形，如右图所示。两个小正方形边上的中点构成阴影多边形的四个顶点，已知阴影四边形的面积为 3，请问这个大正方形未涂阴影区域的面积是多少？

(**A**) 12 (**B**) 15 (**C**) 18 (**D**) 21 (**E**) 24

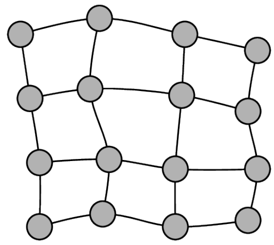
13. What is the greatest common divisor of $2^{2021}+2^{2022}$ and $3^{2021}+3^{2022}$?

$2^{2021}+2^{2022}$ 和 $3^{2021}+3^{2022}$ 的最大公约数是多少？

(**A**) 2^{2021} (**B**) 1 (**C**) 2 (**D**) 6 (**E**) 12

14. The map shows a region with 16 cities connected by roads. The Government wants to build electricity power plants in some of the cities. Each power plant can provide enough electricity for the city where it is sited and any cities connected to that city by a single road. What is the smallest number of power plants that need to be built?

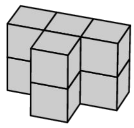
如图表示若干条道路相连的 16 个城市地图，现在政府想在某些城市建造发电厂，每个发电厂可以为所在城市和与该城市通过一条道路连接的其他城市提供充足的电力，请问最少需要建造多少座发电厂？

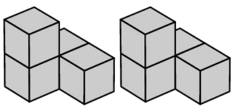


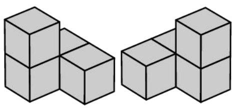
(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

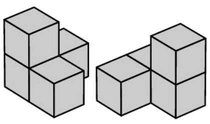
15. Which of the pairs of pieces below can be put together to build the shape shown in the diagram?

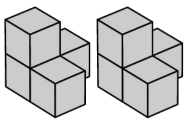
请问以下哪个选项中的两个积木可以组合成右图中的形状？

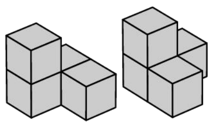


(**A**) 

(**B**) 

(**C**) 

(**D**) 

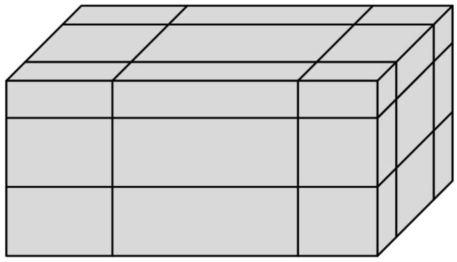
(**E**) 

16. Martina is playing in an 8 player tournament. She knows she will beat everyone except Ash, who will beat everybody. In the first round, players are organised randomly into four pairs, and the winner of each match proceeds to the second round. In the second round, there are two matches and the winners of these matches proceed to the final. What is the probability that Martina does not get to the final?

Martina 正在参加一场 8 人锦标赛。她知道她能打败除了Ash 之外的任何一个人，而 Ash 能打败所有人。比赛第一回合，选手们被随机分成四组，每组的胜方可以进入第二回合，第二回合随机分成两组选手继续比赛，胜方进入决赛。请问 Martina 没有进入决赛的概率是多少？

(**A**) 1 (**B**) 1/2 (**C**) 2/7 (**D**) 3/7 (**E**) 4/7

17. A cuboid of surface area S is cut by six planes as shown. Each plane is parallel to a face, but its distance from the face is random. Now the cuboid is separated in 27 smaller parts. What, in terms of S, is the total surface area of all 27 smaller parts?



将一个表面积为 S 的立方体用 6 个平面切割，每个平面都与立方体的一个表面平行，但距离表面的长度是随机的，如右图所示。这个立方体最后被切割成 27 个小块，请问这 27 个小块的表面积之和是多少（用 S 表示）？

(**A**) $2S$ (**B**) $\frac{5}{2}S$ (**C**) $3S$ (**D**) $4S$

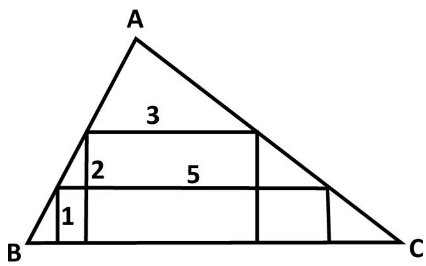
(**E**) none of the previous 以上均不正确

18. Five numbers have a mean of 24. The mean of the three smallest numbers is 19 and the mean of the three largest numbers is 28.
What is the median of the five numbers?

有五个数，平均值是 24，其中三个最小的数的平均值是 19，三个最大的数的平均值是 28。请问这五个数的中位数是多少?

(**A**) 20 (**B**) 21 (**C**) 22 (**D**) 23 (**E**) 24

19. Two rectangles are inscribed inside a triangle ABC. The dimensions of the rectangles are 1×5 and 2×3, respectively, as shown. What is the height of the triangle with base BC?

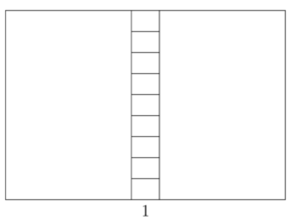


两个尺寸分别为 1×5 和 2×3 的矩形内接于三角形 ABC，如图所示。请问底边为 BC 的三角形的高度是多少?

(**A**) 3 (**B**) $\frac{7}{2}$ (**C**) $\frac{8}{3}$ (**D**) $\frac{16}{5}$

(**E**) none of the previous 以上均不正确

20. A rectangle is divided into 11 smaller rectangles, as shown in the diagram. All 11 rectangles are similar to the original large rectangle. The orientation of the smallest rectangles is the same as the largest. The length of the base of the smallest rectangle is 1. What is the perimeter of the large rectangle?



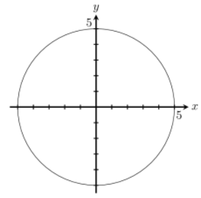
将一个大长方形分成 11 个小长方形，如右图所示。这些小长方形和分割前的大长方形都相似，其中最小长方形和最大长方形的长宽方向一致。已知最小长方形的底边长度为 1，请问大长方形的周长是多少?

(**A**) 20 (**B**) 24 (**C**) 27 (**D**) 30 (**E**) 36

5 points

21. A circle with centre (0, 0) has radius 5. At how many points on the perimeter of the circle are both coordinates integers?

以点 (0, 0) 为圆心的圆的半径为 5，请问圆周上有多少个点的横纵坐标都是整数?



(**A**) 5 (**B**) 8 (**C**) 12 (**D**) 16 (**E**) 20

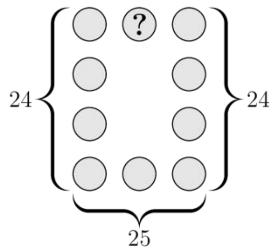
22. How many positive three-digit integers are there that are equal to five times the product of their digits?

有多少个三位正整数等于它的各个数位上数字之积的 5 倍?

(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

23. The numbers 1 to 10 are placed, once each, in the circles of the figure shown. The sum of the numbers in the left column is 24; the sum of the numbers in the right column is also 24 and the sum of the numbers in the bottom row is 25. What number is in the circle containing the question mark?

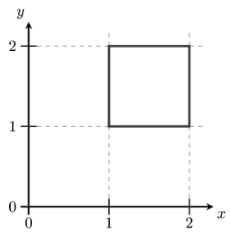
将 1 到 10 放置在右图所示的圆圈中，最左边一列的数字之和是 24，最右边一列的数字之和也是 24，最下面一行的数字之和为 25。请问标记问号的圆圈里应该填哪个数字?

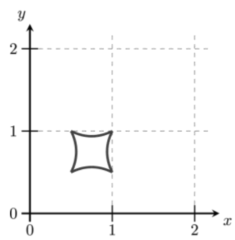


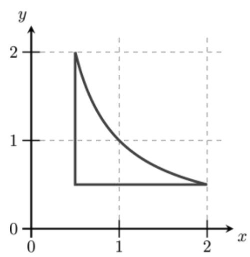
(**A**) 2 (**B**) 4 (**C**) 5
(**D**) 6 (**E**) none of the previous 以上均不正确

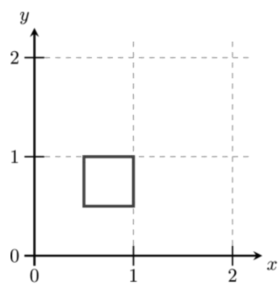
24. A square lies in a coordinate system as shown. Each point (x, y) on the square is moved to $\left(\frac{1}{x}, \frac{1}{y}\right)$. What will the resulting figure look like?

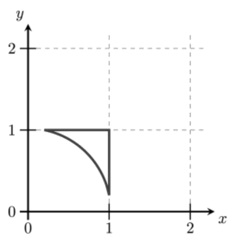
在平面坐标系中，有一个正方形，如右图所示。如果将正方形上的各个点 (x, y) 移到 $\left(\frac{1}{x}, \frac{1}{y}\right)$，请问呈现的最终形状是什么?

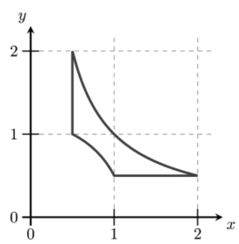


(**A**)


(**B**)


(**C**)


(**D**)


(**E**)


25. The vertices of a 20-gon are numbered from 1 to 20 in such a way that the numbers of adjacent vertices differ by either 1 or 2. The sides of the 20-gon whose ends differ by only 1 are colored red. How many red sides are there?

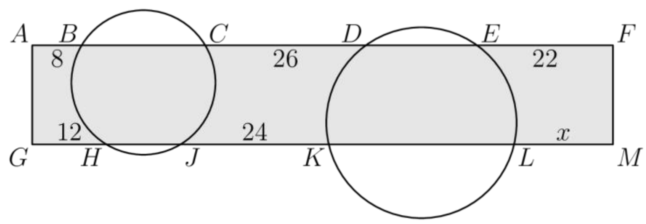
将 1 到 20 标记在一个 20 边形的各个顶点上，使得相邻顶点上的数字之差为 1 或者 2，然后将 20 条边中两端数字只相差 1 的边涂成红色，请问有多少条红色的边?

(**A**) 1 (**B**) 2 (**C**) 5 (**D**) 10

(**E**) there are multiple possibilities 有多种可能性

26. Two circles cut a rectangle $AFMG$, as shown. The line segments outside the circles have length $AB = 8$, $CD = 26$, $EF = 22$, $GH = 12$ and $JK = 24$. What is the length of LM?

两个圆将长方形 $AFMG$ 进行切分，如图所示。 位于圆外的线段长度分别为 $AB = 8$，$CD = 26$，$EF = 22$，$GH = 12$，$JK = 24$。请问 LM 的长度是多少?



(**A**) 14 (**B**) 15 (**C**) 16 (**D**) 17 (**E**) 18

27. Let N be a positive integer. How many integers are there between $\sqrt{N^2+N+1}$ and $\sqrt{9N^2+N+1}$?

已知 N 是一个正整数，请问 $\sqrt{N^2+N+1}$ 和 $\sqrt{9N^2+N+1}$ 之间有多少个整数?

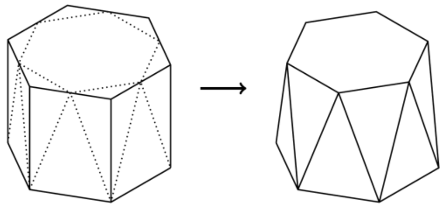
(**A**) $N+1$ (**B**) $2N-1$ (**C**) $2N$ (**D**) $2N+1$ (**E**) $3N$

28. In a sequence, the first term, a_1 is between 0 and 1.
For all $n \geqslant 1$, $a_{2n} = a_2 \cdot a_n + 1$ and $a_{2n+1} = a_2 \cdot a_n - 2$.
Given that $a_7 = 2$, what is the value of a_2?

有一个数列，首项 a_1 介于 0 和 1 之间。
对于所有 $n \geqslant 1$, 满足 $a_{2n} = a_2 \cdot a_n + 1$，$a_{2n+1} = a_2 \cdot a_n - 2$。
如果 $a_7 = 2$, 请问 a_2 的值是多少?

(**A**) Equal to a_1 等于 a_1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

29. A regular hexagonal prism has its top corners shaved off, as shown. The top face becomes a smaller regular hexagon and the 6 rectangular faces around the middle become 12 isosceles triangles of two different sizes. What fraction of the volume of the original prism has been lost?



如图所示，将一个正六棱柱上底面的各个角剃掉后，上底面变成一个更小的正六边形，围绕中心的 6 个矩形侧面变成两种大小不同的 12 个等腰三角形。请问被剃掉的体积占正六棱柱原体积的比例是多少?

(**A**) $\frac{1}{12}$ (**B**) $\frac{1}{6}$ (**C**) $\frac{1}{4\sqrt{3}}$ (**D**) $\frac{1}{6\sqrt{2}}$ (**E**) $\frac{1}{6\sqrt{3}}$

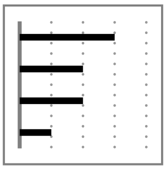
30. A football match between teams from North Berracan and South Berracan is played in a stadium that has a rectangular array of seats for the spectators. There are 11 North Berracan supporters in each row, and 14 South Berracan supporters in each column. This leaves 17 empty seats. What is the smallest possible number of seats in the stadium?

来自北贝拉坎和南贝拉坎的球队正在体育场进行一场足球比赛，体育场的观众席是一个长方形区域。每一排坐着 11 名北贝拉坎球队支持者，每一列坐着 14 名南贝拉坎球队支持者，还有 17 个空座。请问体育场的座位数的最小可能值是多少?

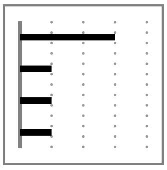
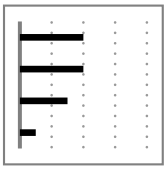
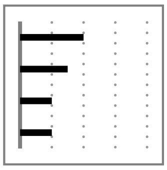
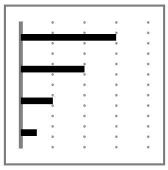
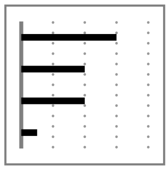
(**A**) 500 (**B**) 660 (**C**) 690 (**D**) 840 (**E**) 994

3 points

1 (DE). On Henry's smartphone, the diagram shows how much time he spent last week on each of his apps.



The apps are ordered from greatest to least time spent. This week, he spent exactly the same amount of time as last week on two of his apps, but only half as much time on the other two. Which of the diagrams below cannot be the diagram for this week?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution: In the diagram in (E), the time spent this week on three apps is equal to the time spent last week.

2 (HK). How many positive three-digit integers are divisible by 13?

(**A**) 68 (**B**) 69 (**C**) 70 (**D**) 76 (**E**) 77

Solution: The smallest three digit number divisible by 13 is $104 = 8 \times 13$, the largest three digit number divisible by 13 is $988 = 76 \times 13$, so there are $76 - 8 + 1 = 69$ multiples of 13 in all three digit numbers.

3 (KR). Bella is older than Charlie and younger than Lily. Teddy is older than Bella. Which two people could be the same age?

(**A**) Charlie and Teddy (**B**) Teddy and Lily (**C**) Lily and Charlie
(**D**) Bella and Lily (**E**) Teddy and Bella

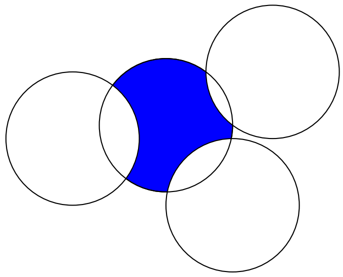
Solution: Lily and Teddy are both older than Bella, while only Charlie is younger.

4 (GR). The product of the digits of a 10-digit integer is 15. What is the sum of the digits of this number?

(**A**) 8 (**B**) 12 (**C**) 15 (**D**) 16 (**E**) 20

Solution: The number can have only one digit 3, one digit 5 and all the other digits must be 1. Asking for a 10-digit number, the digit 1 must be repeated 8 times.

5 (GR). Four circles, each of radius 1, intersect as shown. What is the perimeter of the shaded region?



(**A**) π

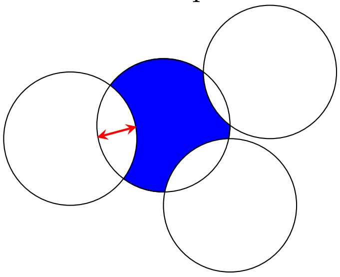
(**B**) Some number between $\frac{3\pi}{2}$ and 2π

(**C**) $\frac{3\pi}{2}$

(**D**) 2π

(**E**) π^2

Solution: It is easy to see that the two arcs between two intersecting circles of the same radii are equal. It follows that the perimeter of the shaded area has perimeter equal to a full circle. So the



total is 2π.

6 (FR). David writes, in increasing order, all the integers from 2 to 2022 which use only 0s and 2s. What is the number in the middle of his list?

(**A**) 200 (**B**) 220 (**C**) 222 (**D**) 2000 (**E**) 2002

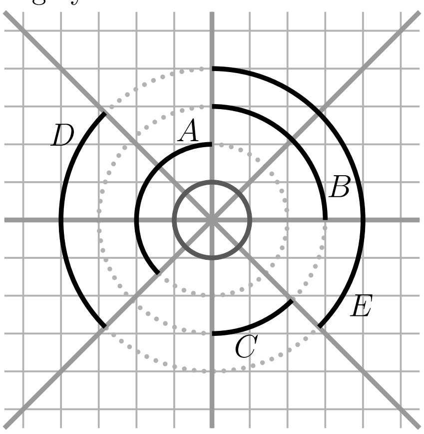
Solution: There are $2^4 = 16$ 4-digit numbers which can be formed using only 0 and 2 (allowing leading zeros). Of these, we do not include 0 and we do not include the numbers greater than 2022 which are 2200, 2202, 2220 and 2222. Thus we have 11 numbers in our list and we are looking for the middle number which will be the sixth in the increasing list. in order, we have 2, 20, 22, 200, 202, 220 so 220 is the number required.

7 (AT). How many real solutions does the equation $(x-2)^2 + (x+2)^2 = 0$ have?

(**A**) 0 (**B**) 1 (**C**) 2 (**D**) 3 (**E**) 4

Solution: Since each term is non-negative, the sum can only be equal to 0 if both are simultaneously equal to 0. Since this is not possible, there are no solutions.

8 (MY). Four lines intersect forming eight equal angles. Which black arc has the same length as the small grey circle?

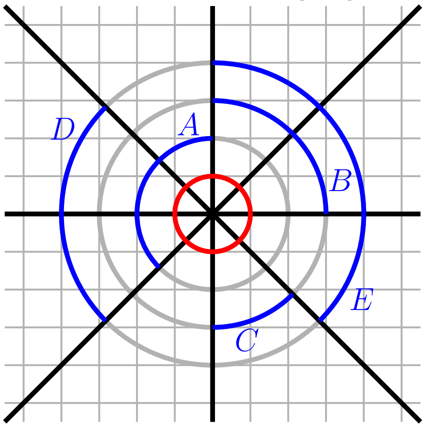


(**A**) A (**B**) B (**C**) C (**D**) D (**E**) E

SOLUTION: Call r the radius of the smallest circle: the other circles have radius $2r$, $3r$, $4r$ respectively. The only arc with lenght $2\pi r$ is D, which corresponds to an angle of $\frac{\pi}{2}$ in a circle of radius $4r$.

Remark For a coloured version of the problem, replace the text with the following:

Four lines intersect forming eight equal angles. Which blue arc has the same length as the red circle?



9 (GR). Let a, b, c be non-zero numbers. The numbers $-2a^4b^3c^2$ and $3a^3b^5c^{-4}$ have the same sign. Which of the following is definitely true?

(**A**) $ab > 0$ (**B**) $b < 0$ (**C**) $c > 0$ (**D**) $bc > 0$ (**E**) $a < 0$

SOLUTION: Two non zero numbers have the same sign if and only if their product is positive. Here the product is $-\frac{6a^7b^8}{c^2}$. Ignoring terms raised to even powers this amounts to $-6a^7 > 0$, and so $a < 0$. Of course we can give examples showing none of the other answers is necessarily true. **Alternative solution**: Since even powers are positive, we get that $-b$ has the same sign of ab: hence $a < 0$, while b and c may be indifferently positive or negative.

10 (CT). Mike has marked the points A, B, C and D in this order on a straight line, as shown in the diagram.

The distance between A and C is 12 cm and between B and D, 18 cm. What is the distance between the midpoint of AB and the midpoint of CD?

(**A**) 15 cm (**B**) 12 cm (**C**) 18 cm (**D**) 6 cm (**E**) 9 cm

SOLUTION: Let $x = \overline{AB}$. Then $\overline{CD} = x+6$. The distance between the two midpoints is $\frac{x}{2}+\overline{BC}+\frac{x}{2}+3 = 15$, since $\overline{BC} + x = \overline{AC} = 12$.

$\boxed{\text{4 points}}$

11 (DE). When he looks at the water meter in his bathroom, Tony notices that all the digits on the meter are different.

$\boxed{\boxed{9}\ \boxed{1}.\boxed{8}\ \boxed{7}\ \boxed{6}\ \text{m}^3}$

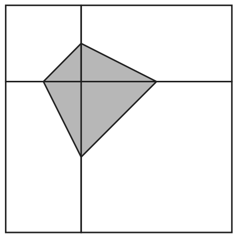
How much water will be used until the next time all the digits on the meter are different?

(**A**) 0.006m^3 (**B**) 0.034m^3 (**C**) 0.086m^3 (**D**) 0.137m^3 (**E**) 1.048m^3

SOLUTION: We cannot increase only the last digit, since it would then be 7, 8 or 9. Analogously we cannot increase only the fourth digit (it would become 8 or 9) or the third (that would become 9).

Hence the second digit must go up to (at least) 2. Choosing the last three digit to obtain the next number in which all digits are different we get 92.013. It appears after using 0.137m^3 of water.

12 (BR). A large square is divided into two unequal squares and two equal rectangles, as shown. The vertices of the shaded quadrilateral are the midpoints of the sides of the two squares. The area of the shaded quadrilateral is 3. What is the area of the unshaded part of the large square?



(**A**) 12 (**B**) 15 (**C**) 18
(**D**) 21 (**E**) 24

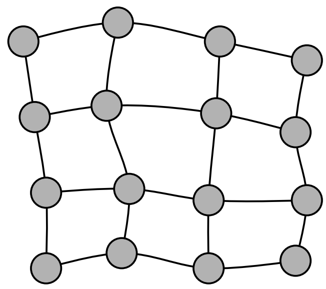
SOLUTION: If $2x$ is the side of the minor square and $2y$ the side of the other, then the shaded region has area $\frac{1}{2} \cdot (x+y)^2 = 3 \iff (x+y)^2 = 6$. The big square has side $2(x+y)$ then it's area is $4 \cdot (x+y)^2 = 4 \cdot 6 = 24$. Therefore, the white part has area $24 - 3 = 21$. **Alternative solution:** Draw the two vertical and the two horizontal lines, not in the picture, that pass through the vertices of the shaded part of the square. Now each of the initial two squares and two rectangles is divided into four equal parts and exactly half of one of these four parts is shaded. So the shaded part is $\frac{1}{8}$ of the big square, and the unshaded part has area $7 \times 3 = 21$.

13 (FR). What is the greatest common divisor of $2^{2021} + 2^{2022}$ and $3^{2021} + 3^{2022}$?

(**A**) 2^{2021} (**B**) 1 (**C**) 2 (**D**) 6 (**E**) 12

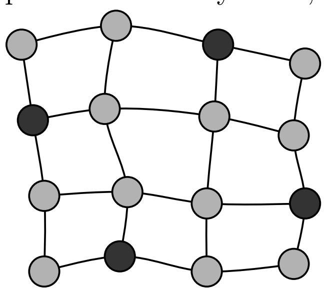
SOLUTION: $2^{2021} + 2^{2022} = 2^{2021} \times 3$ and $3^{2021} + 3^{2022} = 3^{2021} \times 4$.

14 (GR). The map shows a region with 16 cities connected by roads. The Government wants to build electricity power plants in some of the cities. Each power plant can provide enough electricity for the city where it is sited and any cities connected to that city by a single road. What is the smallest number of power plants that need to be built?

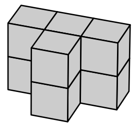


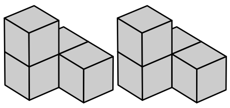
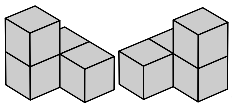
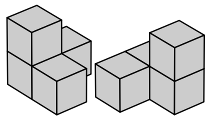
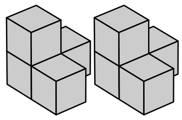
(**A**) 3 (**B**) 4 (**C**) 5
(**D**) 6 (**E**) 7

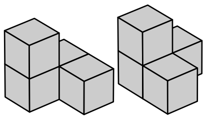
SOLUTION: Each city is connected with at most 4 others, so each factory can provide electricity to at most 5 cities. As 3x5 ¡ 16, it follows that 3 factories are not enough to provide electricity to all, so we need 4 or more. It turns out that 4 are sufficient as seen on the map.



15 (DE). Which of the pairs of pieces below can be put together to build the shape shown in the diagram?



(**A**)  (**B**)  (**C**)  (**D**) 

(**E**) 

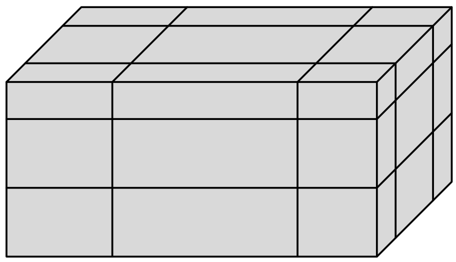
SOLUTION: You can split the original solid shape cutting at middle height the central block (made of 4 small cubes). You obtain two congruent blocks of the shape A.

16 (IT). Martina is playing in an 8 player tournament. She knows she will beat everyone except Ash, who will beat everybody. In the first round, players are organised randomly into four pairs, and the winner of each match proceeds to the second round. In the second round, there are two matches and the winners of these matches proceed to the final. What is the probability that Martina does not get to the final?

(**A**) 1 (**B**) 1/2 (**C**) 2/7 (**D**) 3/7 (**E**) 4/7

SOLUTION: The plan of the matches is random. The second strongest player does not play the final if and only if she falls in the same half-table as the strongest one. In this half-table 3 places are available out of 7.

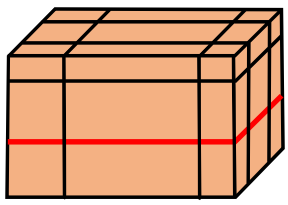
17 (GR). A cuboid of surface area S is cut by six planes as shown. Each plane is parallel to a face, but its distance from the face is random. Now the cuboid is separated in 27 smaller parts. What, in terms of S, is the total surface area of all 27 smaller parts?



(**A**) $2\,S$ (**B**) $\frac{5}{2}\,S$ (**C**) $3\,S$ (**D**) $4\,S$

(**E**) none of the previous

SOLUTION: Let the surface areas of three different faces be A, B, C so that S =2A+2B+2C. Now a cut like the red line shown adds to the total surface area the quantity 2A, where A is the surface area of the face parallel to the cut. Note that the contribution of all six cuts is 4A+4B+4C = 2S, so the total surface area is S+2S=3S.

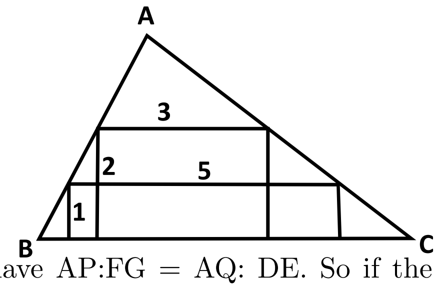


18 (UK). Five numbers have a mean of 24. The mean of the three smallest numbers is 19 and the mean of the three largest numbers is 28.
What is the median of the five numbers?

(**A**) 20 (**B**) 21 (**C**) 22 (**D**) 23 (**E**) 24

SOLUTION: The total of the five numbers is $5 \times 24 = 120$. The total of the three smallest numbers is $3 \times 19 = 57$ and the total of the three largest numbers is $3 \times 28 = 84$. Therefore the middle number is $57 + 84 - 120 = 21$. Hence the median of the five numbers is 21.

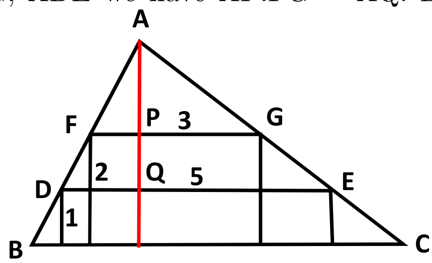
19 (GR). Two rectangles are inscribed inside a triangle ABC. The dimensions of the rectangles are 1×5 and 2×3, respectively, as shown. What is the height of the triangle with base BC?



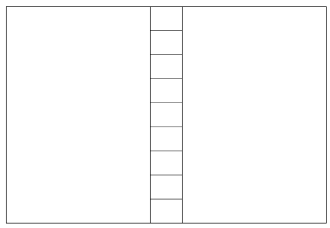
(**A**) 3 (**<u>B</u>**) $\frac{7}{2}$ (**C**) $\frac{8}{3}$ (**D**) $\frac{16}{5}$

(**E**) none of the previous

Solution: From the similarity of the triangles AFG, ADE we have AP:FG = AQ: DE. So if the height is h we have $\frac{h-2}{3} = \frac{h-1}{5}$. Solving we get $h = \frac{7}{2}$.



20 (AU). A rectangle is divided into 11 smaller rectangles, as shown in the diagram. All 11 rectangles are similar to the original large rectangle. The orientation of the smallest rectangles is the same as the largest. The length of the base of the smallest rectangle is 1. What is the perimeter of the large rectangle?

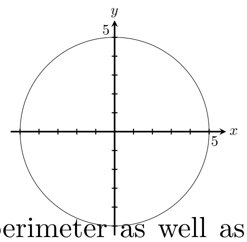


(**A**) 20 (**B**) 24 (**C**) 27 (**<u>D</u>**) 30 (**E**) 36

Solution: The height h of the large rectangle is 9 times the height of the smallest rectangles. As we are told all rectangles are similar, this means the base of the largest rectangle is 9. This means the medium size (upright) rectangles have base 4, so we have $4 : h = h : 9$ which gives $h = 6$. Hence the perimeter of the large rectangle is $2(9 + 6) = 30$.

5 points

21 (FI). A circle with centre $(0, 0)$ has radius 5. At how many points on the perimeter of the circle are both coordinates integers?



(**A**) 5 (**B**) 8 (**<u>C</u>**) 12 (**D**) 16 (**E**) 20

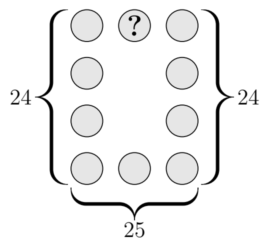
Solution: Since $\sqrt{3^2 + 4^2} = 5$, the points $(\pm 3, \pm 4)$ and $(\pm 4, \pm 3)$ are on the perimeter as well as $(\pm 5, 0)$ and $(0, \pm 5)$, which gives a total of 12 points. Since $5^2 - 1^2$ and $5^2 - 2^2$ are not square numbers, there are no others.

22 (TK). How many positive three-digit integers are there that are equal to five times the product of their digits?

(**<u>A</u>**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

Solution: The last digit must be 5, so the number can only have odd digits. The product of the digits divides by 5, so 5 times this product will divide by 25. The only possibilities are: 175,375,575,775,975. Of these, only 175 satisfies the requirements.

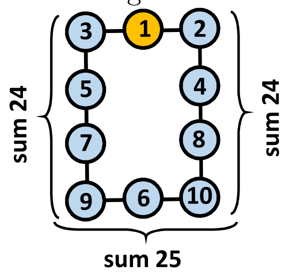
23 (GR). The numbers 1 to 10 are placed, once each, in the circles of the figure shown. The sum of the numbers in the left column is 24; the sum of the numbers in the right column is also 24 and the sum of the numbers in the bottom row is 25. What number is in the circle containing the question mark?



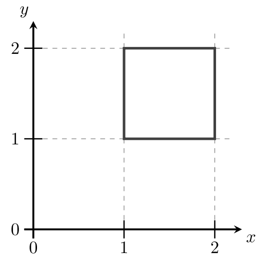
(**A**) 2 (**B**) 4 (**C**) 5

(**D**) 6 (**E**) none of the previous

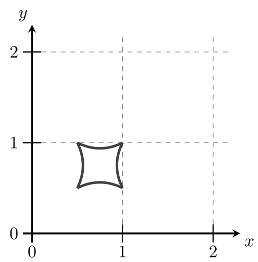
Solution: The maximum possible sum of the numbers in the two columns and the bottom row, where 9 numbers appear of which two are repeated, is $(2+3+4+5+6+7+8+9+10)+(9+10) = 73$. But the sums shown in the figure are $24+25+24 = 73$, which equals this maximum. This is crucial. It means that in these nine circles there is only one choice of numbers, namely 2 to 10 with the numbers 9 and 10 inside the two bottom corners to give the maximum sum. So the number remaining is 1, which goes in the yellow circle. An example to show that the situation is possible is given in the figure.

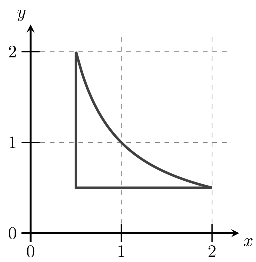


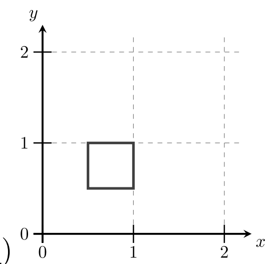
Notice that the two large numbers, 9 and 10, are at the bottom corners.

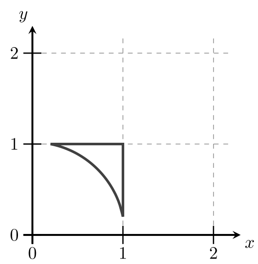


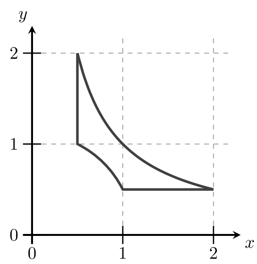
24 (FI). A square lies in a coordinate system as shown.
Each point (x, y) on the square is moved to $\left(\dfrac{1}{x}, \dfrac{1}{y}\right)$. What will the resulting figure look like?

(**A**)


(**B**)


(**C**)


(**D**)


(**E**)


Solution: The corners of the square are mapped like this:

$$(1,1) \mapsto (1,1)$$

$$(1,2) \mapsto (1,1/2)$$

$$(2,1) \mapsto (1/2,1)$$

$$(2, 2) \mapsto (1/2, 1/2)$$

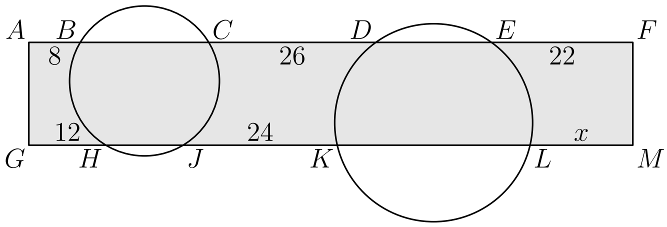
Since all sides of the square have either the x coordinate or the y coordinate constant, the sides will be transformed into horizontal or vertical lines. Therefore the transformed figure will be a square as well, and C is correct.

25 (MX). The vertices of a 20-gon are numbered from 1 to 20 in such a way that the numbers of adjacent vertices differ by either 1 or 2. The sides of the 20-gon whose ends differ by only 1 are colored red. How many red sides are there?

(**A**) 1 (**B**) 2 (**C**) 5 (**D**) 10
(**E**) there are multiple possibilities

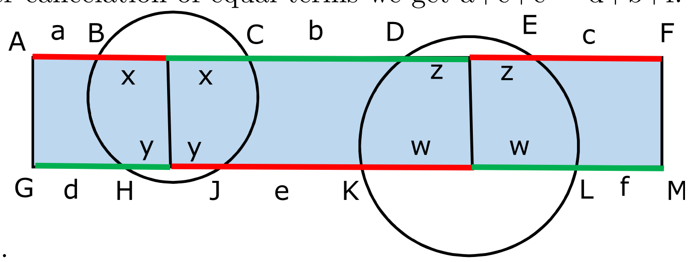
Solution: Numbers 1 and 20 should be opposite in the 20-gon, otherwise, in one of the directions, they would be separated by 9 sides or less, but $1 + 9 \cdot 2 = 19 < 20$. Then the only possible ways to reach 20 in 10 steps starting with 1 are to have even numbers (in order) on one side of the 20-gon and odd numbers on the other side. This leaves a gap of 1 between 1 and 2 and between 19 and 20. All the other gaps are 2.

26 (GR). Two circles cut a rectangle AFMG, as shown. The line segments outside the circles have length AB=8, CD=26, EF=22, GH=12 and JK=24. What is the length of LM?



(**A**) 14 (**B**) 15 (**C**) 16 (**D**) 17 (**E**) 18

Solution: If the given segments are a, b, c, d, e, f we prove, more generally, a+e+c = d+b+f. For this draw the perpendicular bisector of chord BC. It passes through the centre of the circle so it is also the perpendicular bisector of chord HJ. Denote each of the two equal parts of BC by x and of HJ by y. Similarly for DE and KL. Now observe that the sum of the red segments in the figure is equal to the sum of the green segments, so (a+x)+(y+e+w)+(z+c)= (d+y)+(x+b+z)+(w+f). After cancelation of equal terms we get a+e+c = d+b+f. In our case 8+24+22=12+26+LM, so LM



=16.

27 (GR). Let N be a positive integer. How many integers are there between $\sqrt{N^2 + N + 1}$ and $\sqrt{9N^2 + N + 1}$?

(**A**) $N + 1$ (**B**) $2N - 1$ (**C**) $2N$ (**D**) $2N + 1$ (**E**) $3N$

Solution: Note $N < \sqrt{N^2 + N + 1} < N + 1$ and $3N < \sqrt{9N^2 + N + 1} < 3N + 1$. So there are $3N - (N + 1) + 1 = 2N$ integers in between.

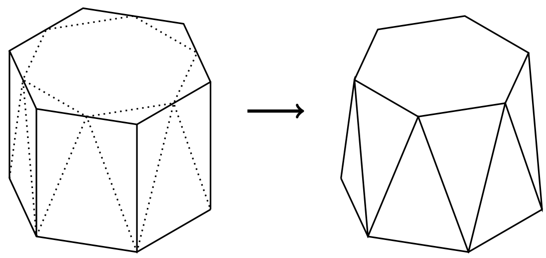
28 (CT). In a sequence, the first term, a_1 is between 0 and 1.
For all $n \geq 1$, $a_{2n} = a_2 \cdot a_n + 1$ and $a_{2n+1} = a_2 \cdot a_n - 2$.
Given that $a_7 = 2$, what is the value of a_2?

(**A**) Equal to a_1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

Solution: We can work backwards in this question. We can see that always $a_{2n} - a_{2n+1} = 3$, and so $a_6 = 5$. From $a_6 = a_3 \cdot a_2 + 1$, we can see that $a_3 \cdot a_2 = 4$ and we also know that $a_2 - a_3 = 3$.

Combining these, we obtain $a_2 = 4$ or $a_2 = -1$. As $a_2 = a_2 \cdot a_1 + 1$ and $0 < a_1 < 1$, it is only possible that $a_2 = 4$.

29 (AU). A regular hexagonal prism has its top corners shaved off, as shown. The top face becomes a smaller regular hexagon and the 6 rectangular faces around the middle become 12 isosceles triangles of two different sizes. What fraction of the volume of the original prism has been lost?



(**A**) $\frac{1}{12}$ (**B**) $\frac{1}{6}$ (**C**) $\frac{1}{4\sqrt{3}}$ (**D**) $\frac{1}{6\sqrt{2}}$ (**E**) $\frac{1}{6\sqrt{3}}$

SOLUTION: Consider the top hexagonal face before and after slicing. From the symmetry of the slicing, the smaller hexagon has its vertices on the midpoints of the sides of the original hexagon. It follows that the original hexagon can be dissected into 24 congruent isosceles 30°-30°-120° triangles, 18 of which cover the smaller hexagon. Therefore the area which is lost is one quarter of the total area.
Each shaved off corner is a pyramid whose base is one of the triangles in the diagram above. Since the volume of a pyramid is one third the volume of the corresponding prism with the same base area and height, and the area lost is one quarter of the total end face, it follows that one twelfth of the volume has been lost overall

30 (AU). A football match between teams from North Berracan and South Berracan is played in a stadium that has a rectangular array of seats for the spectators. There are 11 North Berracan supporters in each row, and 14 South Berracan supporters in each column. This leaves 17 empty seats. What is the smallest possible number of seats in the stadium?

(**A**) 500 (**B**) 660 (**C**) 690 (**D**) 840 (**E**) 994

SOLUTION: Let r denote the number of rows of seats and c the number of columns of seats. Then the total number of seats is rc, which must also be equal to $11r + 14c + 17$. We have

$$rc = 11r + 14c + 17 \Rightarrow (r - 14)(c - 11) = 171.$$

So $r - 14$ and $c - 11$ must be positive integers that multiply to give 171. This reduces the problem to the six cases

$$(r - 14, c - 11) = (1, 171), (3, 57), (9, 19), (19, 9), (57, 3), (171, 1).$$

However, these imply that

$$(r, c) = (15, 182), (17, 68), (23, 30), (33, 20), (71, 14), (185, 12),$$

so

$$rc = 2730, 1156, 690, 660, 994, 2220.$$

Therefore, the smallest possible number of seats in the stadium is 660 and occurs when there are 33 rows and 20 columns. We observe that it is indeed possible to arrange for the spectators to sit in such an arrangement.